

Ali Yavari

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Professional Summary

AI systems researcher with an M.S. in Electrical and Computer Engineering with a 4.0 GPA, focusing on reinforcement learning, time-series prediction, and large-model optimization. Experienced in designing scalable ML architectures across multi-GPU clusters, integrating model-level and system-level improvements for performance and efficiency. Skilled in developing AI pipelines and acceleration strategies using PyTorch, MLflow, and cloud computing platforms. Passionate about advancing AI computing architectures, multimodal optimization, and reinforcement learning.

Education

University of Alberta

Master of Science in Electrical and Computer Engineering

Supervisor: Prof. Qing Zhao – GPA: 4.0/4.0

Sept. 2023 – Jan. 2026
Edmonton, Alberta, Canada

Sharif University of Technology

Bachelor of Science in Mechanical Engineering

Supervisor: Prof. Mehdi Behzad – GPA: 18.13/20

Sept. 2018 – Feb. 2023
Tehran, Iran

Relevant Experience

Machine Learning Resident

Alberta Machine Intelligence Institute (Amii) – In collaboration with Repsol (Spain)

Edmonton, Canada
Aug 2025 – Feb 2026

- Trained large-scale time series forecasting models for complex industrial operations using eXtended LSTMs and Transformers to capture long-range dependencies for data-driven simulation purposes.
- Engineered an MVP for long-horizon forecasting and process-control simulation
- Optimized data ingestion, model parallelism, and checkpointing strategies to enhance training throughput and scalability.
- Collaborated with domain experts to balance predictive accuracy and computational efficiency in long-horizon forecasts.

Graduate Research Assistant

Applied Control & Diagnosis Laboratory, University of Alberta

Edmonton, Canada
Jan. 2024 – Jan 2026

- Applied Safe RL in partially observable markov decision processes(POMDPs) to learn robust control policies, emphasizing constraint satisfaction, and control performance.
- Developed optimal filters via RL and Bayesian Optimization, improving state estimation
- Built evaluation protocols to analyze policy stability, constraint violations, and performance trade-offs

Research Intern

EPFL Excellence in Engineering Program - In collaboration with Hexagon AB

Lausanne, Switzerland
June 2023 – Sept. 2023

- Developed real-time error compensation software in TwinCAT using neural network-based kinematic error modeling for a robot used in meteorology applications.
- Conducted data-driven experiments to optimize motor control performance, reducing cogging torque in PMSMs.
- Integrated low-level system optimization with model-level intelligence to achieve faster and more stable control.

Project Student

Condition Monitoring Laboratory

Tehran, Iran
Jan. 2022 – Jan. 2023

- Developed predictive maintenance models using deep learning for vibration-based fault detection and RUL estimation.
- Engineered hybrid datasets and improved model generalization through data augmentation and precision tuning.

Projects

- **Generative AI Assistant for Technical Debugging (Personal Project)**
 - Developed an LLM-based debugging assistant using OpenAI API and vector retrieval for semantic fault diagnosis in control systems.
 - Integrated multi-modal context retrieval and response generation for domain-specific reasoning tasks.
- **Safe RL for Control in Constrained Environments**
 - Trained PPO agents under safety and performance constraints; developed evaluation metrics for robustness and transferability.
 - Investigated reward shaping and policy regularization for reliable deployment in uncertain environments.
- **Large-Model Forecasting on Industrial Data**
 - Designed and trained multi-layer Transformer models for long-sequence industrial data forecasting with GPU parallelization.
 - Implemented mixed-precision training to balance compute efficiency and numerical stability.

Publications

- Yavari, A., Najafi, A., Zhao, Q., & Jin, J. (2024). Safe Reinforcement Learning via Reward Shaping: A Case Study on Grasping Task in the KUKA Robotic Environment. *Alberta Robotics & Intelligent Systems Expo (ARISE 2024) Award Candidate Paper*.
- Yavari, A., & Zhao, Q. Observer-based control in partially observable environments using Lyapunov-constrained reinforcement learning for estimation and control with performance guarantees. Manuscript submitted.
- Wang, H., Bo, S., Fazeli, S. M., Yavari, A., Liu, B., & Liu, J. Constrained stabilization control via safe reinforcement learning and control Lyapunov value function. Manuscript submitted.

Technical Skills

- **Reinforcement Learning:** Safe RL, POMDPs, Bayesian Optimization, familiarity with RLHF / preference optimization
- **Generative AI & RAG:** Retrieval-Augmented Generation, LLM-based agents, prompt engineering, familiarity with LangChain/agent orchestration patterns.
- **Systems & Acceleration:** GPU/NPU Optimization, Parallel Computing, Mixed Precision (FP16/FP8), Dynamic Scheduling, Model Quantization
- **Programming & Libraries:** Python, PyTorch, BoTorch, PyTorch Lightning, HuggingFace, Scikit-learn, Pandas, NumPy, SQL, MATLAB, C/C++, Git, Bash, unit testing
- **Cloud & MLOps:** Azure, Google Cloud, Docker, Linux, GitHub Actions/Azure DevOps, Weights & Biases

Leadership & Non-Technical Skills

- **Project Ownership:** Led research projects from concept to deployment
- **Collaboration:** Experienced in interdisciplinary research
- **Innovation:** Proactive in exploring emerging AI system architectures

Academic Honors

- **LAB2MARKET (L2M) Program Fellow** – \$10,000 for developing the "SafeRL Engine" commercialization initiative.
- **EPFL E3 Program Fellowship**, Switzerland, 2023 – 3% Acceptance Rate.
- Distinguished Student, Sharif University of Technology, 2022.
- Ranked Top 0.4% among 200,000 University Entrance Exam Participants, 2018.

Relevant Courses

- Robot Learning, Statistical Inference, Generative AI Systems, Advanced Data Analytics, Modern Control, Digital Signal Processing